

NANYANG TECHNOLOGICAL UNIVERSITY

Example Class

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N4-02c-98

School of Computer Engineering

NTU

Course Site: https://ntulearn.ntu.edu.sg/

Schedule

Week	Topic	Tutorial	Lecture 1 (Monday 8:30)	Lecture 2 (Wednesday 8:30)
1	Mod 1 (Intro, Review Verilog, Design Flow, IP)	No Tut	Introduction	Not used
2	Mod 2 (Verification & Verilog Testbenches)	Tut 1	Not Used	Not used
3	Mod 3 (FPGA Architecture and Synthesis)	Tut 2	Not used	Not Used
4	Mod 4 (Arithmetic on FPGAs)	Tut 3	Not used	**EC1** (Mod 1 & 2)
5	Mod 5 (Timing and Pipelining)	Tut 4	Not Used	**EC2** (Mod 3 & 4)
6	Mod 5 (Timing and Pipelining) continued	Tut 5	Not used	Not used
7	Mod 7 (**eLearning** Buses & interfacing)	No Tut	Not used	**QUIZ**
Recess				
8	Mod 8 (uP Case Study)	No Tut	Not Used	**EC3** (Mod 5 & 7)
9	Mod 8 (uP Case Study)	Tut 7	Not used	Not used
10	Mod 9 (Asynchronous Design)	Tut 8	Not used	**EC4** (Mod 8)
11	Mod 9 (Asynchronous Design)	Tut 9	Not Used	Not used
12	--	Tut 10	Not used	**EC5** (Mod 9)
13	--	Not used	Not used	Not used

The yellow shaded regions are for Example Classes (which are in addition to the LAMS-based lectures).
The green shaded region is for a class quiz (details will be provided later).
There is no Tutorial 6 as this has been integrated into the eLearning module.
The current Lab schedule can be accessed from: https://freeaccess.scse.ntu.edu.sg/scselabs

This may change slightly throughout the semester
Individual Tutors are likely to change the tutorial timing.



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Example Class

Module 9 (EC5)

Asynchronous Design

Summary

- Synchronous vs Asynchronous Circuits
- Clocking and clock definitions
- A simple D-Latch
  - Saw that it was not very simple
- Started looking at the Asynchronous design process
  - Used a simple Arbiter as an example
  - Saw there were a number of steps in the design process which was different to the synchronous case
  - Hazards were somewhat problematic
- Hence we needed a more systematic design process

Summary (cont.)

- We examined a more systematic design process
  - Get the flow table and reduce the number of states (if possible)
  - Perform state assignment and derive the excitation table
  - Obtain the next-state and output expressions and the circuit.
- We looked at the Hamming distance, and saw we wanted a square or cubic structure in the state transition diagram.
  - Saw how to eliminate diagonal transitions by adding extra states
  - By rerouting through other temporary states
- Then we looked at a complete example (Example 2).

First, I want to give you a different approach to determining the flow table for example 2.

Example 2 (From Lectures)

Design an asynchronous circuit to the following spec:

- A circuit has 2 inputs, ***I*** and ***Enable***, and an output, ***Q***. If *Enable* is high, a rising edge on *I* causes *Q* to go high. *Q* stays high until *Enable* goes low. **If *Enable* is low, *Q* is low.**
- In the lectures we used a truth table approach to generate the state flow table.

State	I	Enable	Q
A	0	0	0
--	0	0	1
B	0	1	0
C	1	0	0
--	1	0	1
D	1	1	0
E	0	1	1
F	1	1	1

It is also possible to determine a state diagram and generate the state flow table from that.

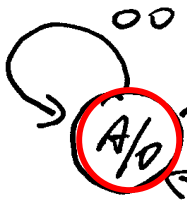
Current State	Next State $IE_n =$ 00	01	11	10	Out Q
A	**A**	B	--	C	0
B	A	**B**	F	--	0
C	A	--	D	**C**	0
D	--	B	**D**	C	0
E	A	**E**	F	--	1
F	--	E	**F**	C	1

Example 2 (Alternative)

Design an asynchronous circuit to the following spec:

– A circuit has 2 inputs, ***I*** and ***Enable***, and an output, ***Q***. If *Enable* is high, a rising edge on *I* causes *Q* to go high. *Q* stays high until *Enable* goes low. If *Enable* is low, *Q* is low.

I En



Start with state A with
IE = 00 & Q = 0.
If E=1 -> B (eg. IE=01).

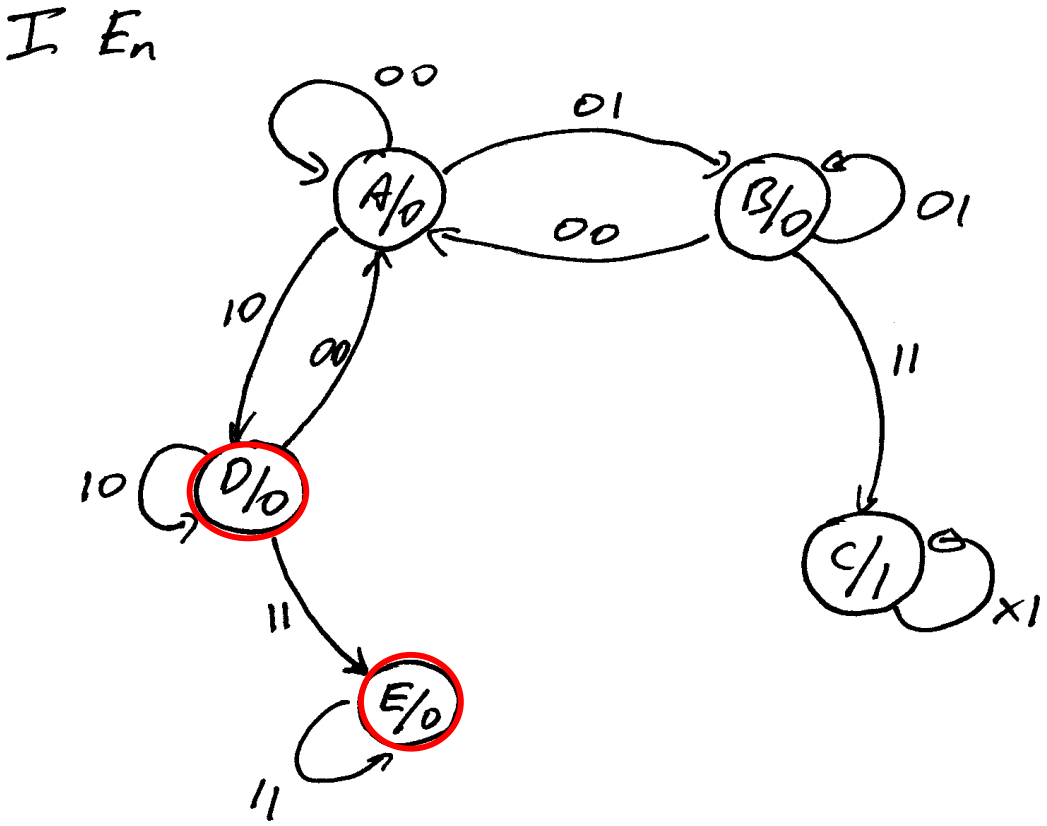
Then if E=0 (eg. IE=00) -> A.
Else if I=1 (eg. IE=11) -> C.

Stay in C if E=1 (eg. IE=X1) .

Example 2 (cont.)

Design an asynchronous circuit to the following spec:

– A circuit has 2 inputs, ***I*** and ***Enable***, and an output, ***Q***. If *Enable* is high, a rising edge on *I* causes *Q* to go high. *Q* stays high until *Enable* goes low. If *Enable* is low, *Q* is low.



If in state A and I=1 -> D (eg. IE=10) .

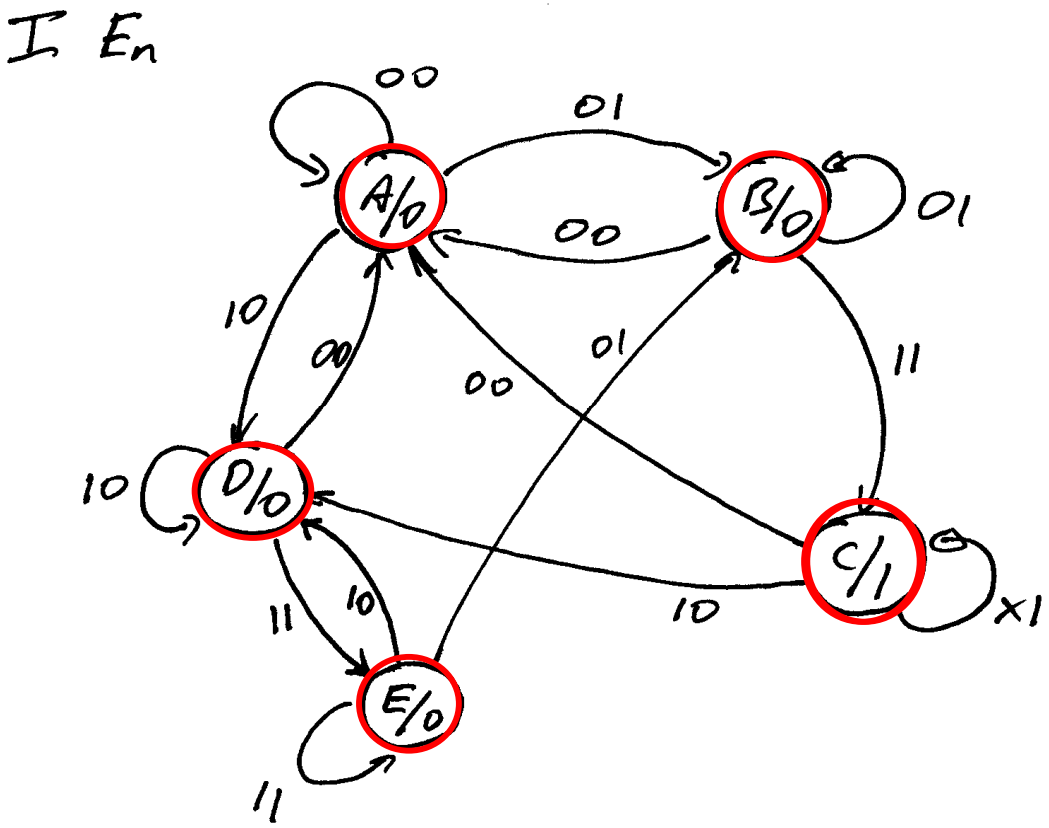
Then if I=0 (eg. IE=00) -> A.

Else if E=1 (e.g. IE=11) -> E.

Example 2 (cont.)

Design an asynchronous circuit to the following spec:

– A circuit has 2 inputs, ***I*** and ***Enable***, and an output, ***Q***. If *Enable* is high, a rising edge on *I* causes *Q* to go high. *Q* stays high until *Enable* goes low. If *Enable* is low, *Q* is low.



If in state E (eg. IE=11) and
E=0 (eg. IE=10) -> D.
Else if I=0 (eg. IE=01) -> B.

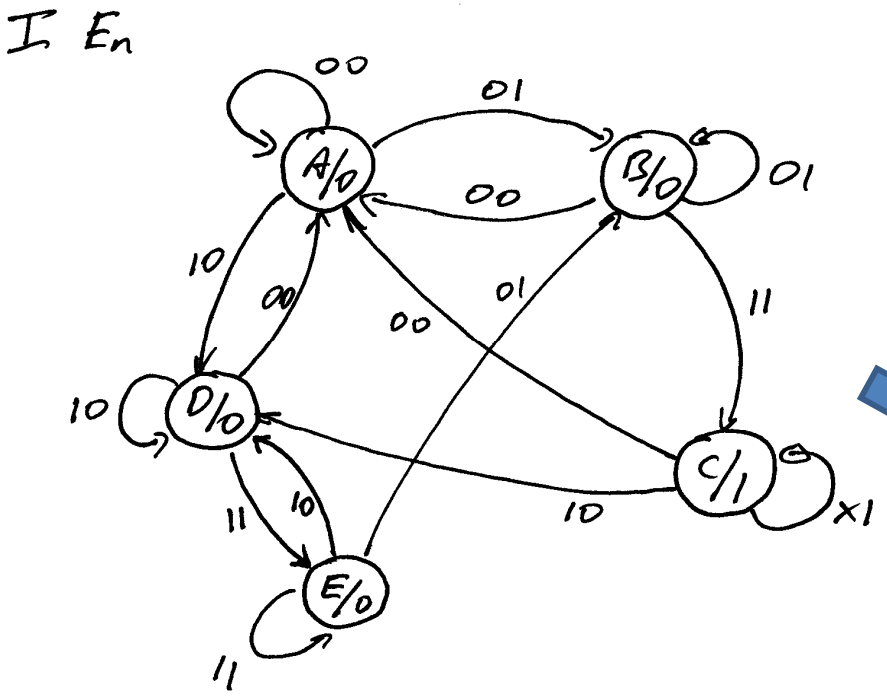
If in state C (with IE=01) and
E->0 (eg. IE=00) -> A.
If in state C (with IE=11) and
E->0 (eg. IE=10) -> D.

So that is the final flow diagram

Example 2 (cont.)

Design an asynchronous circuit to the following spec:

– A circuit has 2 inputs, ***I*** and ***Enable***, and an output, ***Q***. If *Enable* is high, a rising edge on *I* causes *Q* to go high. *Q* stays high until *Enable* goes low. If *Enable* is low, *Q* is low.



Note this looks different from the state table from the lecture notes. For a start, it only has 5 states, the lecture example had 6 states.
So… Simplify by combining states

	I En 00	01	11	10	Q
A	(A)	B	–	D	0
B	A	(B)	C	–	0
C	A	(C)	(C)	D	1
D	A	–	E	(D)	0
E	–	B	(E)	D	0

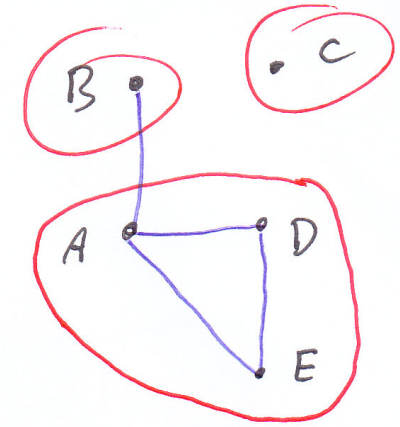
State flow table

Example 2 (cont.)

Design an asynchronous circuit to the following spec:

– A circuit has 2 inputs, ***I*** and ***Enable***, and an output, ***Q***. If *Enable* is high, a rising edge on *I* causes *Q* to go high. *Q* stays high until *Enable* goes low. If *Enable* is low, *Q* is low.

	I En 00	01	11	10	Q
A	(A)	B	–	D	0
B	A	(B)	C	–	0
C	A	(C)	(C)	D	1
D	A	–	E	(D)	0
E	–	B	(E)	D	0



Firstly, get the partitions

P1 = (A B D E) (C)
P2 = ((AB)(AD)(AE)(DE)) (C)

	00	01	11	10	Q
A	(A)	B	(A)	(A)	0
B	A	(B)	C	–	0
C	A	(C)	(C)	A	1

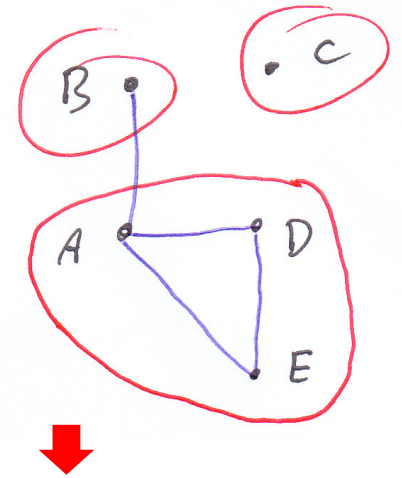
Example 2 (cont.)

Design an asynchronous circuit to the following spec:

– A circuit has 2 inputs, ***I*** and ***Enable***, and an output, ***Q***. If *Enable* is high, a rising edge on *I* causes *Q* to go high. *Q* stays high until *Enable* goes low. If *Enable* is low, *Q* is low.

	IEn 00	01	11	10	Q
A	(A)	B	–	D	0
B	A	(B)	C	–	0
C	A	(C)	(C)	D	1
D	A	–	E	(D)	0
E	–	B	(E)	D	0

P1 = (A B D E) (C)
P2 = ((AB)(AD)(AE)(DE)) (C)



This is the same as in the lecture notes (first grouping), except state C is labelled as state E

EX2 From lectures

Current State	Next State IE = 00	01	11	10	Output Q
A	**A**	B	**A**	**A**	0
B	A	**B**	E	--	0
E	A	**E**	**E**	A	1

	00	01	11	10	Q
A	(A)	B	(A)	(A)	0
B	A	(B)	C	–	0
C	A	(C)	(C)	A	1

Module 9 Quiz Question

What is the best state assignment for the following flow table?

Present State	Next state CD = 00	01	10	11	Output Z_1Z_2
A	**A**	B	C	**A**	00
B	**B**	**B**	D	C	01
C	A	**C**	D	**C**	10
D	B	--	**D**	A	11

We need to check all the possible state assignments. There are 12 possible permutations. But, if we fix one state, say state A to be 00, what we really need to do is check the possible states which are diagonal from A. Remember it is the transitions between diagonal states which cause the dynamic Hazards.

00	A	A	A
01	B	B	C
11	C	D	B
10	D	C	D

There are just 3 possibilities.

B opposite A; C opposite A; or D opposite A

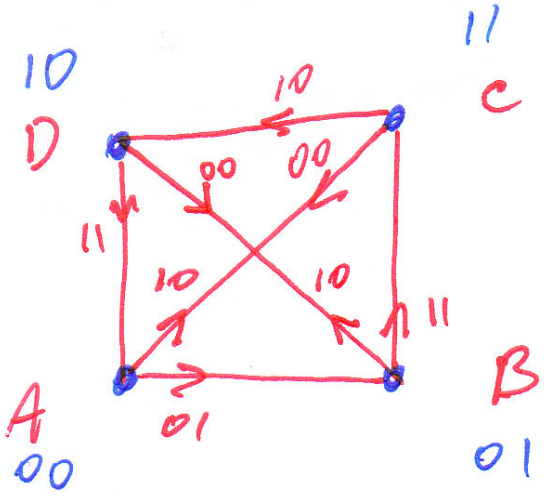
For example, if you take right column: There is no impact to the outcome of the state transition diagram if you take {A, C, B, D} or {A, D, B, C}. It is just flipped.

Note: opposite A = 00 means state 11

Present State	Next state CD = 00	01	10	11	Output Z_1Z_2
A	**A**	B	C	**A**	00
B	**B**	**B**	D	C	01
C	A	**C**	D	**C**	10
D	B	--	**D**	A	11

00	A	A	A
01	B	B	C
11	C	D	B
10	D	C	D

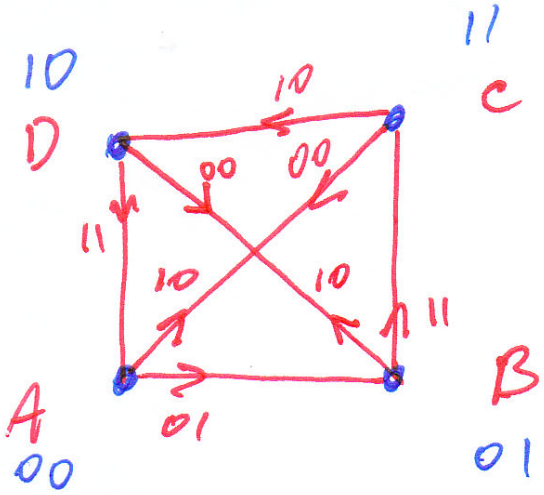
Then we can generate the state transition diagrams as:

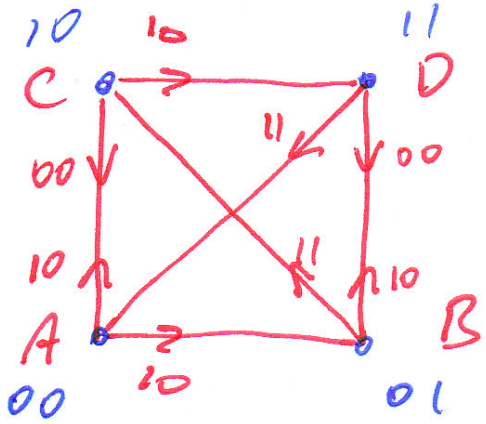


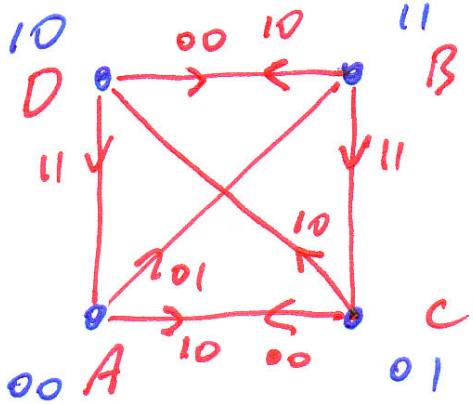
Present State	Next state CD = 00	01	10	11	Output Z_1Z_2
A	**A**	B	C	**A**	00
B	**B**	**B**	D	C	01
C	A	**C**	D	**C**	10
D	B	--	**D**	A	11

00	A	A	A
01	B	B	C
11	C	D	B
10	D	C	D

Then we can generate the state transition diagrams as:







Note that the first has 4 diagonal transitions, so it is not a solution, but the others are possibilities.

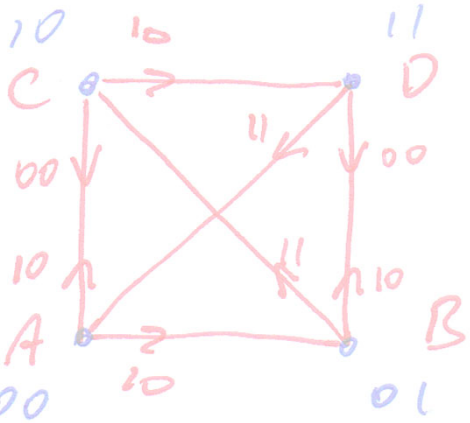
Present State	Next state CD = 00	01	10	11	Output Z_1Z_2
A	**A**	~~B~~ D	~~C~~ D	**A**	00
B	**B**	**B**	D	C	01
C	A	**C**	~~D~~ A,B	**C**	10
D	B	~~-~~ B	**D**	A	11

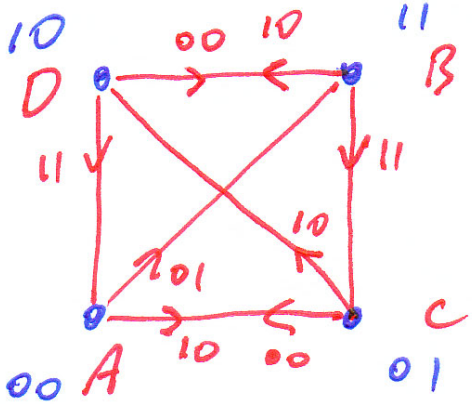
00	A	A	A
01	B	B	C
11	C	D	B
10	D	C	D

ANS:

Looking at the left case, we see that for input 11, we need to move from state B to state C. But we cannot use either state D or state A as temporary states.

So it is not a solution.





So this just leaves the right case. The first diagonal transition is state A to B on input 01. Here we can use the undefined state in D and transition A->D->B. The other diagonal transition is state C to D on input 10. We could transition from C to B and then to stable D, or we could change state A to go directly to D.

Exam: Solutions

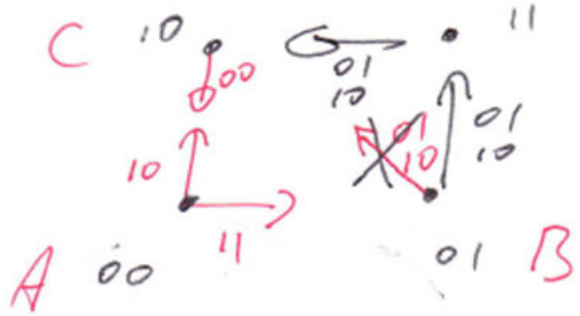
AY1819 Q3(b) You are required to implement the **asynchronous** circuit, described by the flow table of Table Q3. Using the state assignment {A=00, B=01 and C=10}, sketch a **hazard free** gate level implementation of the circuit using just AND and OR gates. Bubble inputs should be used if required.

Table Q3

Present State	Input XY = 00	01	11	10	Output (Z)
A	**A**	**A**	B	C	0
B	--	C	**B**	C	1
C	A	**C**	**C**	**C**	0

Note: a bold and underlined state represents a stable state

First, check the state transitions.



We see that there are 2 diagonal transitions from state B to state C (in red).

So add a redundant state D (11) and route both through D (in black). Note it is also possible to redirect B=10 through state A.

Exam: Solutions

AY1819 Q3(b)

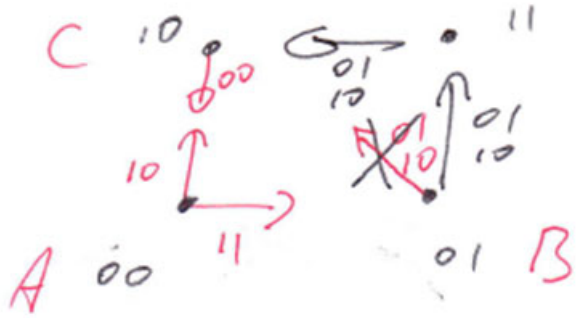
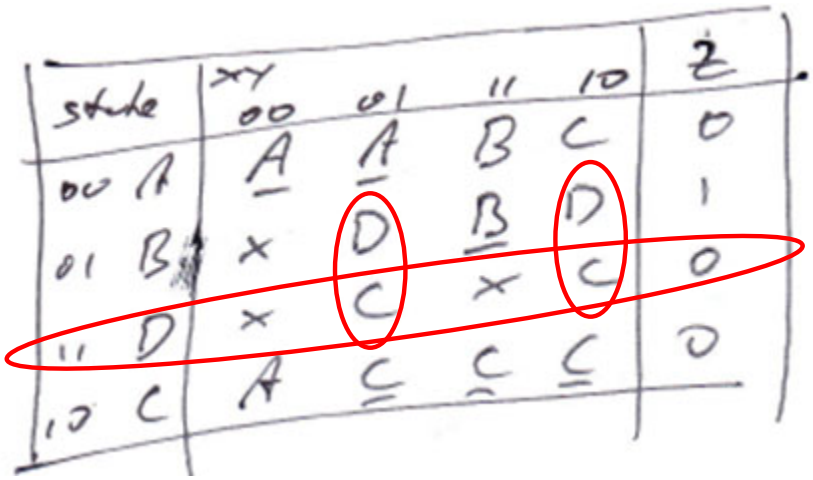


Table Q3

Present State	Input XY = 00	01	11	10	Output (Z)
A	**A**	**A**	B	C	0
B	--	C	**B**	C	1
C	A	**C**	**C**	**C**	0

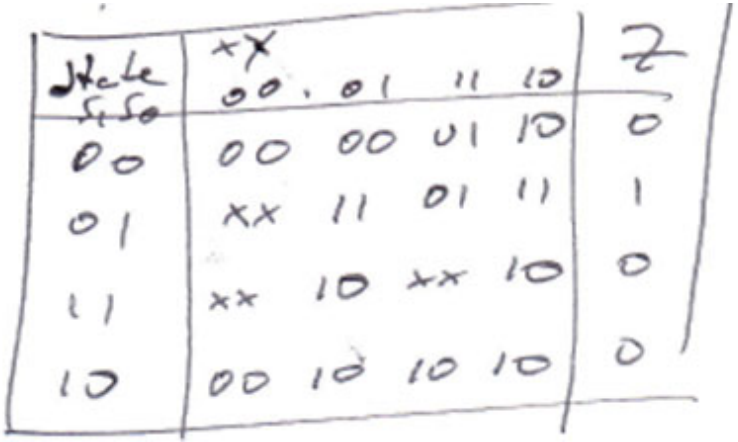
Note: a bold and underlined state represents a stable state

The state table and the excitation table:



state	xy 00	01	11	10	z
00 A	A	A	B	C	0
01 B	x	D	B	D	1
11 D	x	C	x	C	0
10 C	A	C	C	C	0

State table

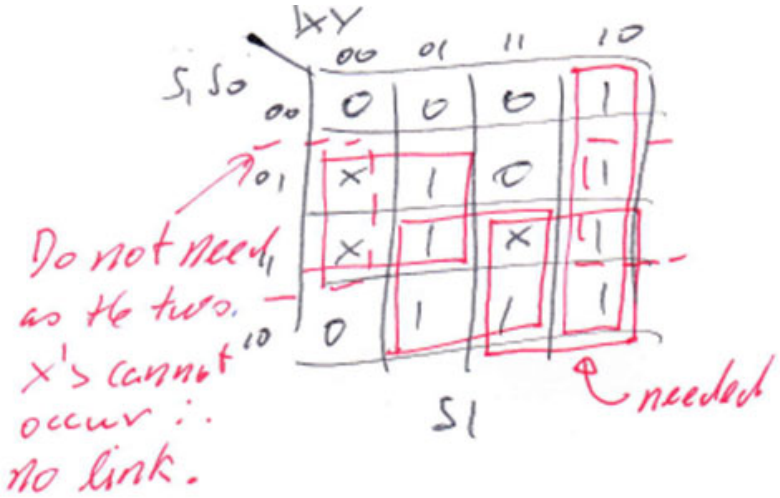


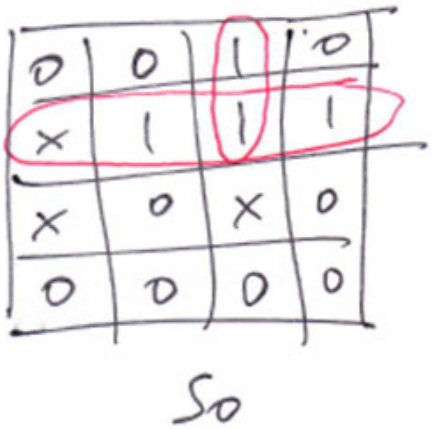
state y_1y_0	xx 00	01	11	10	z
00	00	00	01	10	0
01	xx	11	01	11	1
11	xx	10	xx	10	0
10	00	10	10	10	0

Excitation table

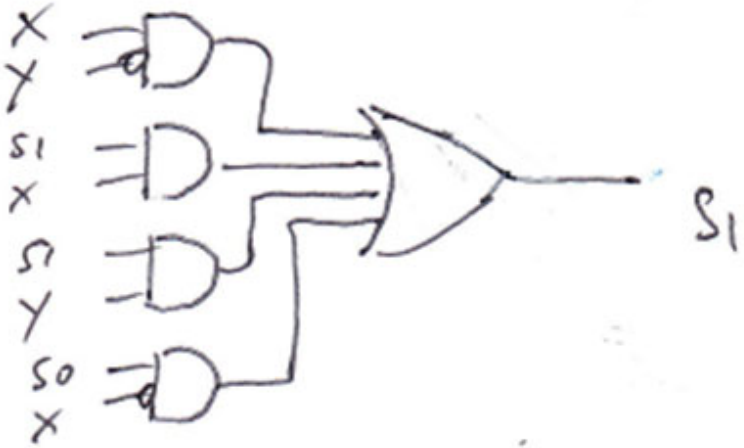
Exam: Solutions

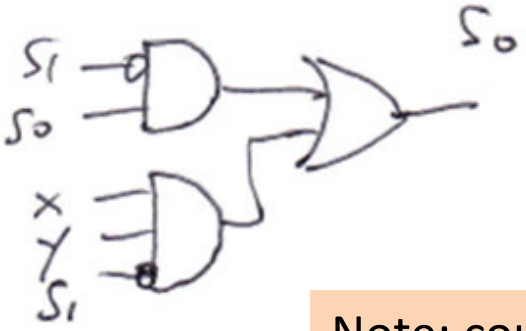
AY1819 Q3(b) Then the K-maps:





$S_1^+ = XY' + S_1X + S_1Y + S_0X'$ $S_0^+ = S_1'S_0 + XYS_1'$ $Z = S_0S_1'$







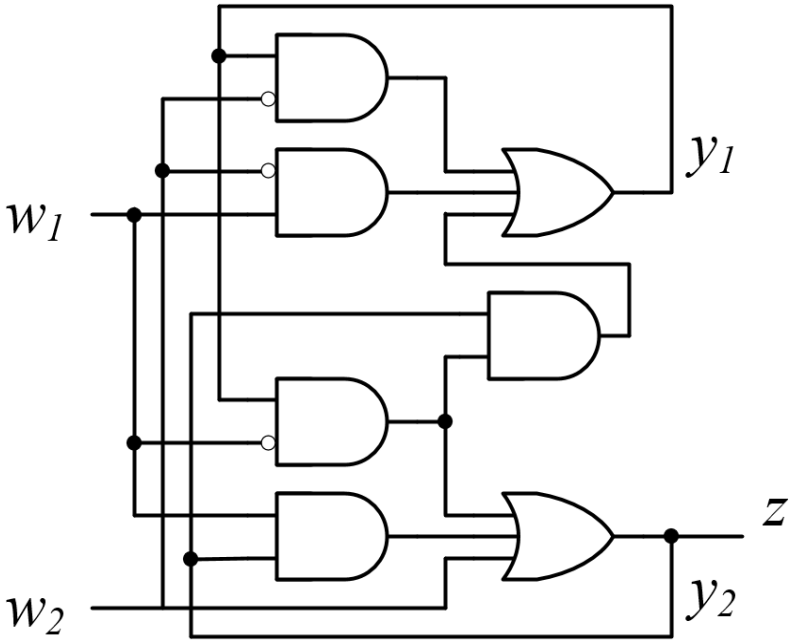
Note: could set Z=1 for state D that would simplify Z to just S_0

AY1718 Exam: Solutions

Q4. The figure shows an implementation of an asynchronous circuit with two inputs (w_1 and w_2), two state variables (y_1 and y_2) and one output (z).

(a) Identify a static hazard and dynamic hazard in the implementation.

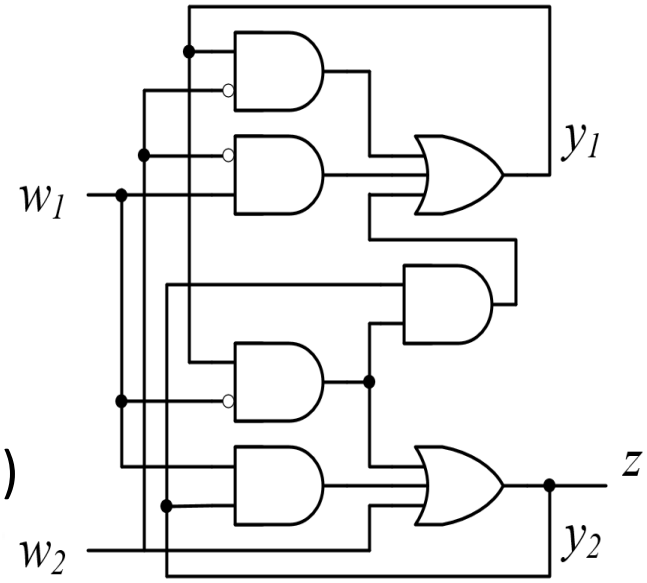
(14 marks)



AY1718 Exam: Solutions

Q4. The figure shows an implementation of an asynchronous circuit with two inputs (w_1 and w_2), two state variables (y_1 and y_2) and one output (z).

(a) Identify a static hazard and dynamic hazard in the implementation.



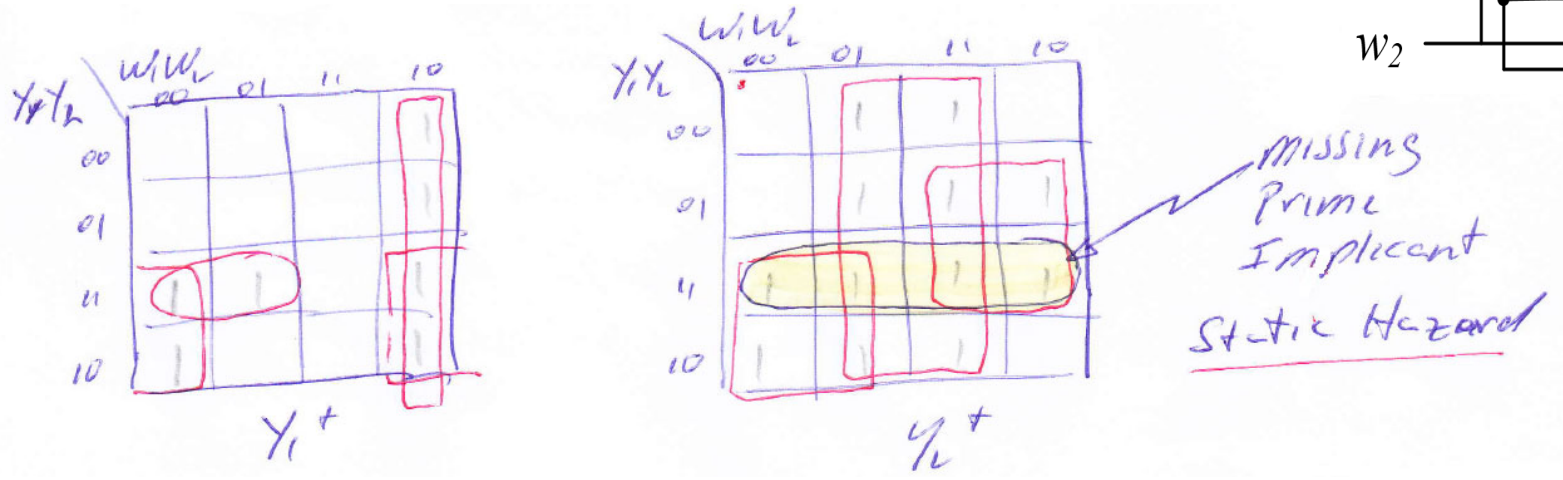
First. Determine the next state and output expressions

$$Y_1^+ = y_1 \cdot \bar{w}_2 + w_1 \cdot \bar{w}_2 + y_2 \cdot y_1 \cdot \bar{w}_1$$

$$Y_2^+ = w_2 + y_2 w_1 + y_1 \bar{w}_1$$

$$z = y_2$$

Next, determine the K-map (from next state expressions)



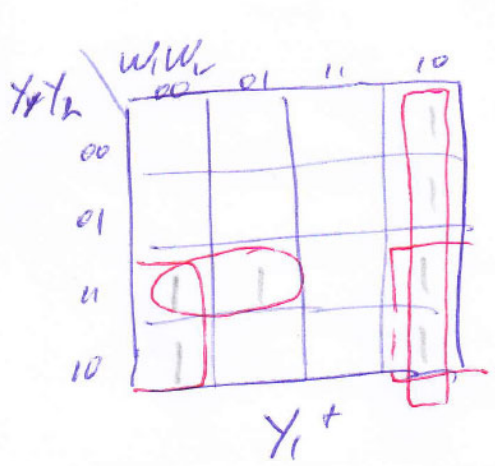
AY1718 Exam: Solutions

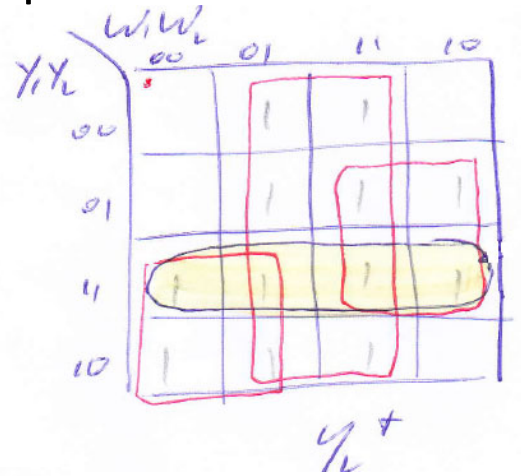
Q4. The figure shows an implementation of an asynchronous circuit with two inputs (w_1 and w_2), two state variables (y_1 and y_2) and one output (z).

(a) Identify a static hazard and dynamic hazard in the implementation.

Next, determine the execution table

Y_1Y_2	w_1w_2 = 00	01	11	10	z
00	00	01	01	10	0
01	00	01	01	11	1
11	11	11	01	11	1
10	11	01	01	10	0





Then the flow table is:

	00	01	11	10	z
A	A	B	B	D	0
B	A	B	B	C	1
C	C	C	B	C	1
D	C	B	B	D	0

don't care

don't care due to input change

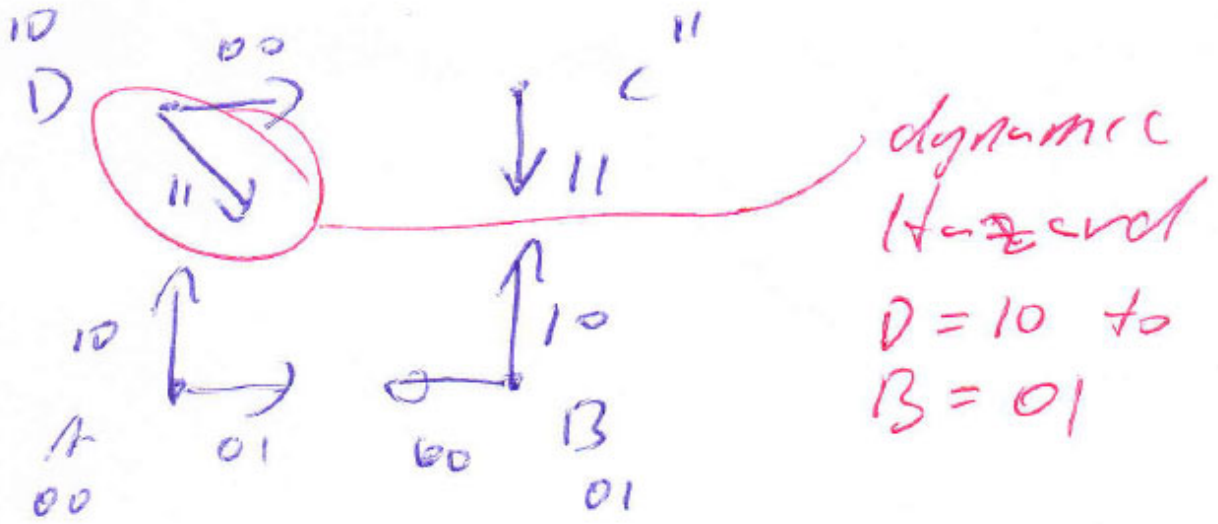
But note that a row with just a single stable state will actually have don't cares (only a 1 bit input change is allowed).

AY1718 Exam: Solutions

Q4. The figure shows an implementation of an asynchronous circuit with two inputs (w_1 and w_2), two state variables (y_1 and y_2) and one output (z).

(a) Identify a static hazard and dynamic hazard in the implementation.

Hence, draw the state transition diagram.
(excluding the 2 don't care states)



	00	01	11	10	
A	A	B	~~B~~	D	0
B	A	B	B	C	1
C	C	C	B	C	1
D	C	~~B~~	B	D	0

don't care

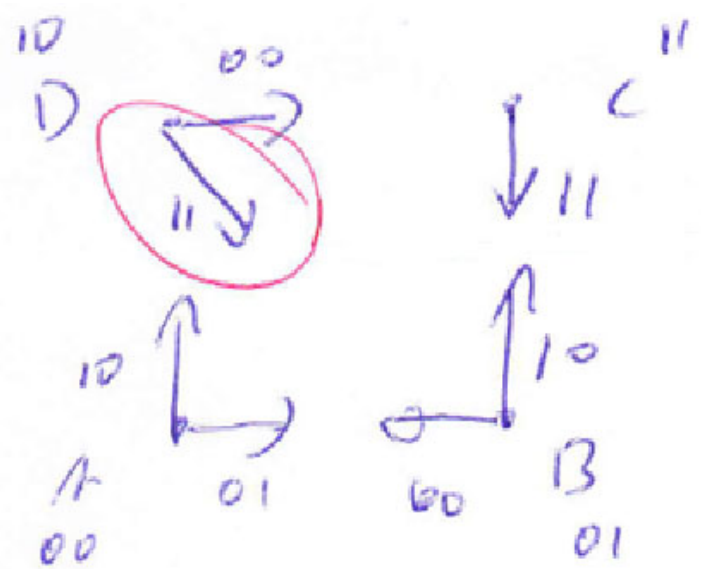
don't care due to input change

The static hazard is because **ALL prime implicants** were not included in the next state expression for Y2. The dynamic hazard is because the transition from state D to state B has a **diagonal transi**tion (eg. 2 bits change).

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Q4. (b) Determine the hazard free excitation table and hence the next state expressions. You do not need to draw the gate level implementation or reproduce the output expression. (11 marks)

We firstly need to eliminate the diagonal transition from state D to state B.



	00	01	11	10	
A	A	B	×	D	0
B	A	B	B	C	1
C	C	C	B	C	1
D	C	×	~~D~~ C	D	0

It is fairly obvious that we can transition from state D to C to B (with input 11)

Could have also gone through state A (don't care)

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Q4. (b) Determine the hazard free excitation table and hence the next state expressions. You do not need to draw the gate level implementation or reproduce the output expression.

Hence the excitation table is .

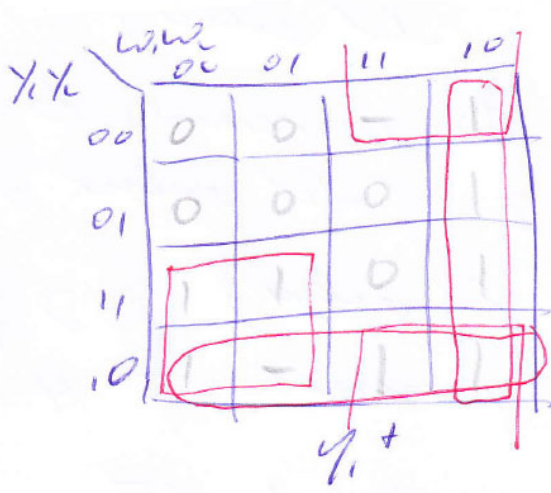
Y_1Y_2	w_1w_2 = 00	01	11	10	z
00	00	01	--	10	0
01	00	01	01	11	1
11	11	11	01	11	1
10	11	--	11	10	0

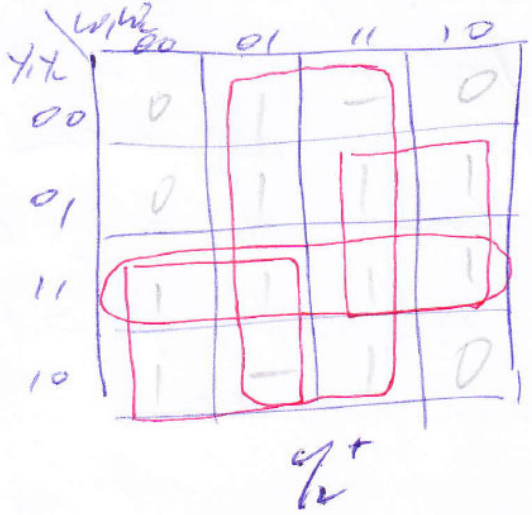
	00	01	11	10	
A	A	B	×	D	0
B	A	B	B	C	1
C	C	C	B	C	1
D	C	×	~~X~~ C	D	0

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Q4. (b) Determine the hazard free excitation table and hence the next state expressions. You do not need to draw the gate level implementation or reproduce the output expression.

Then the K-maps are;





Y_1Y_2	W_1W_2 = 00	01	11	10	z
00	00	01	--	10	0
01	00	01	01	11	1
11	11	11	01	11	1
10	11	--	11	10	0

BUT… Is this correct?

And hence the next state expressions are:

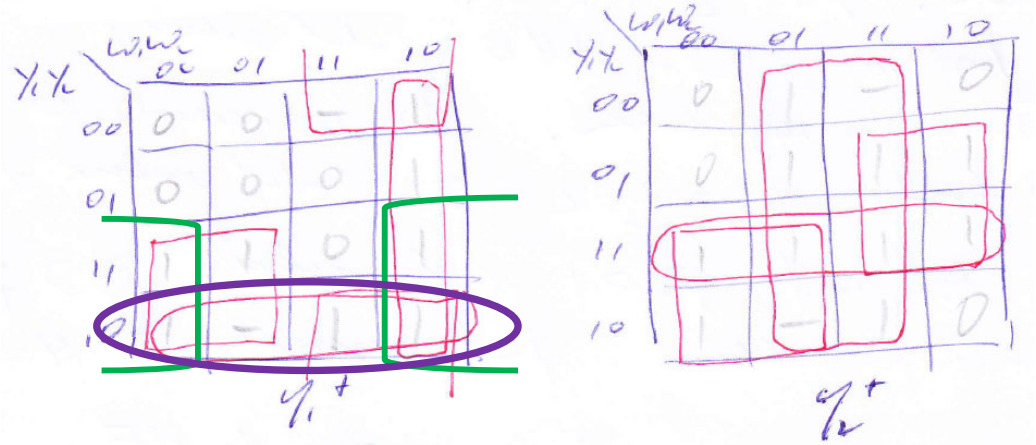
$$Y_1^+ = W_1\overline{W_2} + Y_1\overline{Y_2} + Y_1\overline{W_1} + \overline{Y_2}W_1$$

$$Y_2^+ = W_2 + Y_1Y_2 + Y_1\overline{W_1} + Y_2W_1$$

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But, there is an error in the previous implementation.

All prime implicants (PI) are not included in y_1

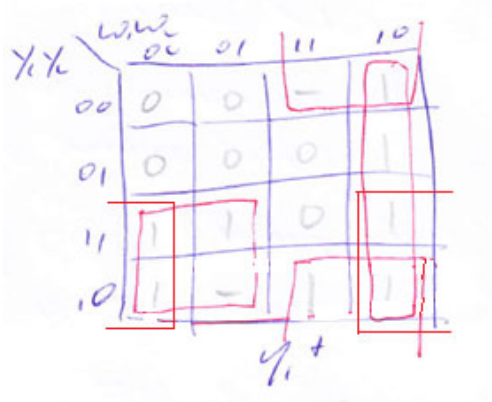


Y_2 is unchanged, but Y_1 becomes:

$$Y_1^+ = W_1W_2' + Y_1Y_2' + Y_1W_1' + W_1Y_2' + Y_1W_2'$$

However, there are other better solutions (as we have some non essential PIs)

1. Don't need Y_1Y_2' as there can be NO transition from 1001 (don't care) to 1011



2. Or don't need W_1Y_2' as there is no need to include the don't care at 0011

